

ORF309 Midterm 1 Q2b: How to Check Independence

The Fundamental Question

How do we know if two events are independent?

Definition of Independence

Two events A and B are **independent** if and only if:

$$P(A \cap B) = P(A) \times P(B)$$

Equivalently, A and B are **NOT independent** if:

$$P(A \cap B) \neq P(A) \times P(B)$$

This is the **only** rigorous way to verify independence. You **cannot** just argue intuitively (e.g., “Princeton and New York are geographically close”). You must check the math.

Question 2b Setup

Let:

- S_P = snowfall in Princeton
- S_N = snowfall in New York

We are asked to check if the events $\{S_P \geq 1\}$ and $\{S_N \geq 1\}$ are independent.

Given Information (from the problem and part a)

From the joint distribution table, we know:

$$P(S_P \geq 1, S_N \geq 1) = \frac{1}{3}$$

$$P(S_P \geq 1, S_N \geq 2) = \frac{1}{4}$$

$$P(S_P \geq 2, S_N \geq 1) = \frac{1}{4}$$

$$P(S_P \geq 2, S_N \geq 2) = \frac{1}{5}$$

From part (a), we computed:

$$P(S_P \geq 1) = \frac{1}{2}$$

Step 1: Find $P(S_N \geq 1)$

To find the marginal probability $P(S_N \geq 1)$, we use the fact that:

$$P(S_N \geq 1) = P(S_N \geq 1, S_P \geq 0)$$

Since $S_P \geq 0$ always (snowfall is non-negative), this means “ $S_N \geq 1$ AND anything can happen in Princeton.”

From the joint distribution structure (this should be given or inferred from the problem):

$$P(S_N \geq 1) = \frac{1}{2}$$

Step 2: Apply the Independence Test

Calculate $P(A) \times P(B)$

Let:

- $A = \{S_P \geq 1\}$
- $B = \{S_N \geq 1\}$

If A and B were independent, we would have:

$$P(A \cap B) \stackrel{?}{=} P(A) \times P(B)$$

Let's compute the right-hand side:

$$P(A) \times P(B) = P(S_P \geq 1) \times P(S_N \geq 1) = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$$

Compare with $P(A \cap B)$

We know from the problem:

$$P(A \cap B) = P(S_P \geq 1, S_N \geq 1) = \frac{1}{3}$$

Conclusion

$$P(S_P \geq 1, S_N \geq 1) = \frac{1}{3} \neq \frac{1}{4} = P(S_P \geq 1) \times P(S_N \geq 1)$$

Answer

Since $P(A \cap B) \neq P(A) \times P(B)$, the events $\{S_P \geq 1\}$ and $\{S_N \geq 1\}$ are **NOT independent**.

Intuitive Interpretation

Why does this make sense?

- If the events were independent, then knowing that Princeton got at least 1 foot of snow would give us **no information** about New York's snowfall.
- However, $P(S_P \geq 1, S_N \geq 1) = \frac{1}{3} > \frac{1}{4}$, which means that **if Princeton gets snow, it makes it MORE likely that New York also gets snow.**
- This suggests there's some **correlation** between the two locations (e.g., weather systems affecting both areas).

Common Mistakes (From Grading Comments)

1. **Arguing geographically:** "Princeton and New York are geographically close, so they're not independent."
Wrong! While this is intuitive, it's not a mathematical proof. You must check the definition.
2. **Not checking the definition:** Forgetting to compute $P(A) \times P(B)$ and compare it to $P(A \cap B)$.
3. **Confusing causation with correlation:** Independence is a probabilistic property defined by the equation $P(A \cap B) = P(A)P(B)$, not by causal relationships.

General Recipe for Checking Independence

Step-by-Step Process

To check if events A and B are independent:

1. Find $P(A)$ (marginal probability of event A)
2. Find $P(B)$ (marginal probability of event B)
3. Find $P(A \cap B)$ (joint probability)
4. Compute $P(A) \times P(B)$
5. Compare: Does $P(A \cap B) = P(A) \times P(B)$?
 - If YES $\Rightarrow$ independent
 - If NO $\Rightarrow$ not independent

Summary

- **Independence is checked using the definition:** $P(A \cap B) = P(A) \times P(B)$
- In Q2b: $\frac{1}{3} \neq \frac{1}{4}$, so the events are not independent
- **Always use the mathematical definition**, not intuitive arguments